

claude-windows / a9fd4426-f234-4e4a-bfb4-1f2c8255c661

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-muneem\`a9fd4426-f234-4e4a-bfb4-1f2c8255c661\subagents\workflows\wf_076beba8-59c\agent-aa04a1ef4e1069ac1.jsonl`  
- SHA-256: `aae06bdf4562c39845ff0a00efd50bc3053013cad6187f67a942cbcb5cf52c2a`  
- Source modified: `2026-05-31T04:24:30+00:00`  
- Imported at: `2026-07-05T16:48:20+00:00`  
- project: `wf_076beba8-59c`  
- session_id: `a9fd4426-f234-4e4a-bfb4-1f2c8255c661`
```

Transcript

1. user (2026-05-31T04:23:18.001Z)

Adversarial Claim Verifier (voter 1/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Identify the best LOCAL (self-hosted, single NVIDIA CUDA GPU) vision / document-AI / OCR models ? and, as a clearly-flagged approval-gated fallback only, paid cloud API services ? for parsing BANK STATEMENT PDFs into structured transaction tables (date, narration/description, withdrawal/debit, deposit/credit, running balance), for a LOCAL-FIRST personal-finance tool ("muneem").

Target documents: Indian bank statements (HDFC, Axis, ICICI, AU Small Finance) ? dense, multi-column, often SEMI-RULED or borderless tables, sometimes multiple accounts per statement; both born-digital (text layer present) and the harder cases where pdfplumber's extract_tables/coordinate clustering is unreliable (e.g. Axis, whose column layout varies between statements).

HARD CONSTRAINTS (a model that violates these is unusable for us):

1. MUST run 100% LOCALLY for statement contents (highly sensitive). NO cloud APIs by default. Paid/cloud is acceptable ONLY as an explicitly-approved fallback ? still report the best cloud options but label them clearly as "cloud/sensitive ? approval-gated, not default".
2. MUST be usable from Python (transformers / vLLM / Ollama / ONNX Runtime / native lib).
3. License MUST be permissive & commercially safe for BOTH the CODE and the MODEL WEIGHTS: MIT, BSD, Apache-2.0, ISC, HPND. AVOID and explicitly FLAG anything GPL/LGPL/AGPL, CC-BY-NC / non-commercial, research-only, or custom "community"/acceptable-use licenses with restrictions. This is critical ? many document-AI models and their training datasets are non-commercial (e.g. LayoutLMv3 weights are CC-BY-NC; Surya and Marker use GPL-or-commercial dual licensing; several VLM weights ship custom restrictive licenses). Verify the WEIGHTS license, not just the repo license.
4. Hardware: a single consumer/prosumer NVIDIA GPU. Give VRAM tiers (~8-12 GB consumer, ~16-24 GB) and which models/quantizations fit each.

Evaluate and compare these LOCAL approaches ? for each give: code license, MODEL-WEIGHTS license (state explicitly), latest release / maintenance status as of 2025-2026, VRAM need + quantization options, Python integration path, and reliability specifically on DENSE FINANCIAL TABLES / multi-column statements:

- Open vision-language models (VLMs) for document/table understanding: Qwen2-VL and Qwen2.5-VL (note per-size license ? which sizes are Apache-2.0 vs restricted), MiniCPM-V 2.6, InternVL2 / 2.5, Microsoft Phi-3.5-vision, Mistral Pixtral, Meta Llama-3.2-Vision (flag the Llama Community License restrictions), Moondream2, GOT-OCR2.0, olmOCR and other OCR-specialized VLMs.
- OCR-specialist / document-AI pipelines: PaddleOCR + PP-Structure (table recognition), docTR, EasyOCR, Tesseract (already used), Microsoft Table Transformer / TATR (check code AND PubTables-1M dataset/weights license), Donut, Surya (FLAG license), Marker (FLAG license),

MinerU / PDF-Extract-Kit, RapidOCR.

- Table-structure recognition specifically (cell/grid reconstruction for borderless tables):
TATR, PP-Structure, UniTable, etc.

Also assess: for SEMI-RULED / borderless bank tables, is a prompt-driven VLM (page image -> JSON transactions) more reliable than a classic OCR + table-structure pipeline? What are the accuracy/hallucination risks of VLMs on financial NUMBERS (transposed digits, dropped/duplicated rows), and how should output be VERIFIED? (muneem already machine-checks parses by balance reconciliation ? opening + sum(deposits) - sum(withdrawals) == closing, plus per-row running-balance walks and cross-checks against the bank's printed summary ? so any model's output can be automatically validated, which should inform the recommendation.)

Finally, briefly cover the best PAID/CLOUD fallbacks (clearly flagged approval-gated for sensitive data): AWS Textract (AnalyzeExpense / Tables), Google Document AI, Azure AI Document Intelligence, Reducto, LlamaParse, Anthropic/OpenAI vision ? note any self-hosting option and data-handling terms.

Deliver a RANKED shortlist of the most promising LOCAL options for muneem (NVIDIA GPU, Python, commercial-license-safe), each with license (code + weights), VRAM tier, maturity, strengths/weaknesses for dense bank-statement tables, and a concrete recommendation: which local model(s) to prototype for muneem's statement parsing and how they'd combine with the existing deterministic parsers + balance-reconciliation verification (e.g. VLM as a fallback/cross-check when deterministic parsing fails to reconcile). Cite sources and call out any license uncertainty explicitly.

Claim under review

"On olmOCR-Bench the model scores 82.3% overall with 84.3% on Tables and 84.3% on Multi-column layouts ? directly relevant to dense, multi-column bank-statement tables."

****Source:**** <https://huggingface.co/allenai/olmOCR-2-7B-1025> (primary)

****Supporting quote:**** "Tables | 84.3% ... Multi-column | 84.3% ... Overall | 82.3 ± 1.1%"

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-05-31T04:23:20.799Z)

I'll verify this claim by examining the primary source and searching for any contradicting evidence.

3. assistant (2026-05-31T04:23:23.154Z)

I'll fetch the primary source to verify the exact benchmark numbers claimed.

4. user (2026-05-31T04:23:31.367Z)

olmOCR-Bench Scores for olmOCR-2-7B-1025

Based on the model card, here are the exact benchmark scores:

Overall Score

82.3 ± 1.1 (for olmOCR-2-7B-1025 with olmOCR pipeline v0.4.0)

Per-Category Scores (olmOCR-2-7B-1025)

Category	Score
ArXiv	82.9
Old Scans Math	82.1
Math	84.3
Tables	84.3
Old Scans	48.3
Headers and Footers	95.7
Multi column	84.3
Long tiny text	81.4
Base	99.7

Model Specification

These scores are for **THIS specific model** (olmOCR-2-7B-1025 in BF16 format) when used with **olmOCR pipeline v0.4.0**.

The card also notes that the FP8 quantized version (olmOCR-2-7B-1025-FP8) achieves nearly identical scores (82.4 ± 1.1 overall), and recommends using the FP8 version for practical purposes except for further fine-tuning.

No specific date is provided on the card, but it references the model as a recent release fine-tuned from Qwen2.5-VL-7B-Instruct with GRPO RL training.

5. user (2026-05-31T04:23:34.491Z)

Web search results for query: "olmOCR-2-7B-1025 olmOCR-Bench overall 82.3 tables multi-column score"

Links: [{"title": "olmOCR 2: Unit test rewards for document OCR | A12", "url": "https://allenai.org/blog/olmocr-2"}, {"title": "allenai/olmOCR-2-7B-1025-FP8 · Hugging Face", "url": "https://huggingface.co/allenai/olmOCR-2-7B-1025-FP8"}, {"title": "OlmOCR-Bench: OCR Evaluation Benchmark", "url": "https://www.emergentmind.com/topics/olmocr-bench"}, {"title": "richardyoung/olmOCR-2-7B-1025-MLX-8bit · Hugging Face", "url": "https://huggingface.co/richardyoung/olmOCR-2-7B-1025-MLX-8bit"}, {"title": "allenai/olmOCR-2-7B-1025 · Hugging Face", "url": "https://huggingface.co/allenai/olmOCR-2-7B-1025"}, {"title": "richardyoung/olmOCR-2-7B-1025-MLX-6bit · Hugging Face", "url": "https://huggingface.co/richardyoung/olmOCR-2-7B-1025-MLX-6bit"}, {"title": "olmOCR-2-7B-1025 OCR Model", "url": "https://www.emergentmind.com/topics/olmocr-2-7b-1025"}, {"title": "[2510.19817] olmOCR 2: Unit Test Rewards for Document OCR", "url": "https://arxiv.org/abs/2510.19817"}, {"title": "olmOCR 2: Unit Test Rewards for Document OCR", "url": "https://arxiv.org/pdf/2510.19817"}]

Based on the search results, I found information about olmOCR-2-7B-1025 and its performance metrics:

olmOCR-2-7B-1025 Performance Overview

olmOCR-2-7B-1025 achieves state-of-the-art performance for real-world OCR of English-language digitized print documents. On olmOCR-Bench, olmOCR 2 achieves 82.4 points, a nearly +4-point improvement over the previous release and one of the highest scores to date.

Specific Benchmark Scores

According to the Hugging Face model card, the model scores 82.4 overall, 83 on ArXiv Math, 82.3 on Old Scans Math, 84.9 on Table Tests, 96.1 on Multi Column on the olmOCR-bench evaluation.

Key Strengths

olmOCR-2-7B-1025 achieves improvements of +19.7 (ArXiv), +14.8 (Old Scans Math), +22.6 (Tables), +16.1 (Multi-column), and +27.1 points across various categories.

The model was fine-tuned from Qwen2.5-VL-7B-Instruct using the olmOCR-mix-1025 dataset and additionally fine-tuned using GRPO RL training to boost its performance at math equations, tables, and other tricky OCR cases.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-05-31T04:23:38.334Z)

I'm noticing a discrepancy. The WebFetch of the primary source confirms the quoted numbers (Overall 82.3 ± 1.1, Tables 84.3, Multi column 84.3), but the WebSearch surfaces a competing set of numbers (Overall 82.4, Tables 84.9, Multi Column 96.1). Let me dig into this discrepancy, since it affects whether the claim's specific numbers are accurate, and check the table-row interpretation more carefully.

7. user (2026-05-31T04:23:46.551Z)

Benchmark Results Table Analysis

The table structure is:

Model	ArXiv	Old Scans Math	Tables	Old Scans Headers and Footers	Multi column	Long tiny text	Base	Overall
olmOCR pipeline v0.4.0 with olmOCR-2-7B-1025	82.9	82.1	84.3	48.3	95.7	84.3	81.4	99.7
olmOCR pipeline v0.4.0 with olmOCR-2-7B-1025-FP8	83.0	82.3	84.9	47.7	96.1	83.7	81.9	99.7

Key Findings:

- The 96.1 value: 96.1 belongs to the "Multi column" column and is associated with olmOCR-2-7B-1025-FP8 (the FP8 version), NOT the BF16 version
- The 84.3 values: 84.3 for "Tables" is associated with olmOCR-2-7B-1025 (the BF16 version) 84.3 for "Long tiny text" is also associated with olmOCR-2-7B-1025 (the BF16 version) So 84.3 belongs specifically to the olmOCR-2-7B-1025 BF16 model, appearing in two different columns
- Multi column row values: olmOCR-2-7B-1025: 95.7 olmOCR-2-7B-1025-FP8: 96.1
- Overall row values: olmOCR-2-7B-1025: 82.3 ± 1.1 olmOCR-2-7B-1025-FP8: 82.4 ± 1.1

8. user (2026-05-31T04:23:47.902Z)

olmOCR-Bench Scores Extraction

Based on the benchmark table in the content, here are the exact scores:

olmOCR pipeline v0.4.0 with olmOCR-2-7B-1025

- Overall: 82.3 ± 1.1

- **Tables**: 84.3
- **Multi column**: 84.3

olmOCR pipeline v0.4.0 with olmOCR-2-7B-1025-FP8

- **Overall**: 82.4 ± 1.1
- **Tables**: 84.9
- **Multi column**: 83.7

Answer to Your Questions

Is there a value of 96.1 anywhere? Yes - that's the **Headers and Footers** score (96.1) for the FP8 variant.

Is 84.9 for Tables? Yes - the FP8 variant scores **84.9 on Tables** (compared to 84.3 for the base model).

Are Tables and Multi column both 84.3? Only for the base olmOCR-2-7B-1025 model. For the FP8 variant, they differ: Tables = 84.9, Multi column = 83.7.

9. assistant (2026-05-31T04:23:52.378Z)

The two WebFetch calls are giving me inconsistent readings of the same table ? the column alignment is ambiguous when the page is converted to markdown (the "Old Scans" / "Headers and Footers" columns seem to be merging or shifting, throwing off the alignment). This is exactly the kind of thing I must not get wrong. I need to read the raw table structure directly rather than rely on a model's interpretation of converted markdown. Let me fetch the raw HTML/README and also check the arXiv paper, which is the authoritative source.

10. user (2026-05-31T04:23:59.061Z)

I don't have access to the olmOCR-Bench results table with the specific scores you're asking about. The webpage content provided only includes the abstract and metadata for the paper, not the detailed results table.

To find the exact scores for olmOCR-2-7B-1025 on Overall, Tables, and Multi-column layouts, you would need to view the full PDF or HTML version of the paper, which contains the benchmark results table.

11. user (2026-05-31T04:24:00.125Z)

olmOCR-Bench Scores Table

Here is the raw markdown table exactly as written:

```
...
<table>
  <thead>
    <tr>
      <th align="left"><strong>Model</strong></th>
      <th align="center">ArXiv</th>
      <th align="center">Old Scans Math</th>
      <th align="center">Tables</th>
      <th align="center">Old Scans</th>
      <th align="center">Headers and Footers</th>
      <th align="center">Multi column</th>
      <th align="center">Long tiny text</th>
      <th align="center">Base</th>
      <th align="center">Overall</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td align="left">olmOCR pipeline v0.4.0 with olmOCR-2-7B-1025</td>
```

```

      <td align="center">82.9</td>
      <td align="center">82.1</td>
      <td align="center">84.3</td>
      <td align="center">48.3</td>
      <td align="center">95.7</td>
      <td align="center">84.3</td>
      <td align="center">81.4</td>
      <td align="center">99.7</td>
      <td align="center">82.3 ± 1.1</td>
    </tr>
    <tr>
      <td align="left">olmOCR pipeline v0.4.0 with olmOCR-2-7B-1025-FP8</td>
      <td align="center">83.0</td>
      <td align="center">82.3</td>
      <td align="center">84.9</td>
      <td align="center">47.7</td>
      <td align="center">96.1</td>
      <td align="center">83.7</td>
      <td align="center">81.9</td>
      <td align="center">99.7</td>
      <td align="center">82.4 ± 1.1</td>
    </tr>
  </tbody>
</table>
...

```

****olmOCR-2-7B-1025 data row:****

```

| olmOCR pipeline v0.4.0 with olmOCR-2-7B-1025 | 82.9 | 82.1 | 84.3 | 48.3 | 95.7 | 84.3 | 81.4
| 99.7 | 82.3 ± 1.1 |

```

12. assistant (2026-05-31T04:24:16.787Z)

Now I have the raw HTML table, which is definitive. The columns map as follows for olmOCR-2-7B-1025 (BF16):

```

| Column | Value |
|---|---|
| ArXiv | 82.9 |
| Old Scans Math | 82.1 |
| **Tables** | **84.3** |
| Old Scans | 48.3 |
| Headers and Footers | 95.7 |
| **Multi column** | **84.3** |
| Long tiny text | 81.4 |
| Base | 99.7 |
| **Overall** | **82.3 ± 1.1** |

```

This is the raw HTML source of the primary model card, so the column-to-value mapping is unambiguous. The claim's three numbers ? Overall 82.3, Tables 84.3, Multi-column 84.3 ? are all exactly correct for the olmOCR-2-7B-1025 (BF16) model. The earlier confusion (96.1, 84.9) came from the FP8 variant row and from markdown-conversion misalignment, not from the actual claim.

Now let me apply skeptical scrutiny to the *interpretation* and *relevance* part of the claim, not just the raw numbers.

****Checklist assessment:****

1. ****Supported by quote?**** Yes. The quote "Tables | 84.3% ... Multi-column | 84.3% ... Overall | 82.3 ± 1.1%" matches the raw HTML table for olmOCR-2-7B-1025 exactly. Verified against the canonical README source.

2. ****Contradicting evidence?**** The only apparent contradiction (96.1 multi-column, 84.9 tables) turned out to be the ****FP8 variant's**** numbers (and one was actually the Headers-and-Footers column), not a dispute of the claim. No credible source disputes the BF16 figures.

3. ****Source quality?**** This is the primary source (the official Ai2/allenai HuggingFace model card), appropriate for the claim's strength. The arXiv paper (2510.19817) backs the same model.

4. ****Outdated?**** The model is olmOCR-2-7B-1025 (Oct 2025 release per the "1025" tag and arXiv date 2510). Current as of the May 2026 research window.

5. ****Marketing/cherry-picked?**** The numbers themselves are accurately reported. However ? and this is the one substantive caveat ? the claim appends the editorial gloss *****directly relevant to dense, multi-column bank-statement tables.***** This is an inferential leap. olmOCR-Bench's "Tables" and "Multi-column" categories are general document benchmarks (academic papers, scanned docs); they are ****not**** a measure of accuracy on dense Indian bank-statement transaction tables with running balances. A strong score on olmOCR-Bench Tables does not establish reliability on financial numerics (transposed digits, dropped rows) ? the exact risk the research question flags. But this relevance gloss is a softer, reasonable framing ("directly relevant" = topically applicable), not a fabricated empirical claim.

The ****factual core**** of the claim ? the three benchmark numbers and that they're olmOCR-2-7B-1025's ? is fully verified against the primary source. The relevance framing is defensible (table/multi-column performance is genuinely the relevant axis), even if it shouldn't be read as proof of bank-statement reliability.

Verdict: ****not refuted.**** The numbers are exact, current, and from the primary source. The "directly relevant" phrasing is a mild, reasonable inference rather than an overreach that would falsify the claim.

13. user (2026-05-31T04:24:26.034Z)

Structured output provided successfully

14. assistant (2026-05-31T04:24:30.055Z)

I've completed the adversarial verification. The claim's three benchmark numbers (Overall 82.3 ± 1.1, Tables 84.3, Multi-column 84.3) were verified exactly against the raw README HTML of the primary source (allenai/olmOCR-2-7B-1025), with the column-to-value mapping confirmed unambiguously. My refutation attempt ? a competing number set (82.4/84.9/96.1) found in search ? turned out to be the FP8 variant's row, not a real contradiction. Verdict: ****not refuted****, high confidence, with a flagged caveat that the "directly relevant to bank-statement tables" gloss is a reasonable inference rather than proof of financial-numeric reliability.